# MaxUnpool3d#

https://pytorch.org/docs/stable/generated/torch.nn.modules.pooling.MaxUnpool

**Description:**

Computes a partial inverse of MaxPool3d. MaxPool3d is not fully invertible, since the non-maximal values are lost. MaxUnpool3d takes in as input the output of MaxPool3d including the indices of the maximal values and computes a partial inverse in which all non-maximal values are set to zero. This operation may behave nondeterministically when the input indices has repeat values. See pytorch/pytorch#80827 and Reproducibility for more information.

## Function Signature

```
class torch.nn.modules.pooling.MaxUnpool3d(kernel_size,
stride=None, padding=0)[source]#

Parameters

Inputs:

Shape:

forward(input, indices, output_size=None)[source]#
```

**Returns:**

Tensor

## Code Examples

```
>>> # pool of square window of size=3, stride=2
>>> pool = nn.MaxPool3d(3, stride=2, return_indices=True)
>>> unpool = nn.MaxUnpool3d(3, stride=2)
>>> output, indices = pool(torch.randn(20, 16, 51, 33, 15))
>>> unpooled_output = unpool(output, indices)
>>> unpooled_output.size()
torch.Size([20, 16, 51, 33, 15])
```

## Notes & Warnings

**NOTE** Note This operation may behave nondeterministically when the input indices has repeat values. See pytorch/pytorch#80827 and Reproducibility for more information.

**NOTE** Note MaxPool3d can map several input sizes to the same output sizes. Hence, the inversion process can get ambiguous. To accommodate this, you can provide the needed output size as an additional argument output_size in the forward call. See the Inputs section below.

# Detailed Documentation

## Documentation

Computes a partial inverse of MaxPool3d.

MaxPool3d is not fully invertible, since the non-maximal values are lost. MaxUnpool3d takes in as input the output of MaxPool3d including the indices of the maximal values and computes a partial inverse in which all non-maximal values are set to zero.

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MaxPool3d can map several input sizes to the same output sizes. Hence, the inversion process can get ambiguous. To accommodate this, you can provide the needed output size as an additional argument output_size in the forward call. See the Inputs section below.

kernel_size (int or tuple) – Size of the max pooling window.

stride (int or tuple) – Stride of the max pooling window. It is set to kernel_size by default.

padding (int or tuple) – Padding that was added to the input

input: the input Tensor to invert

indices: the indices given out by MaxPool3d

output_size (optional): the targeted output size

Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

or as given by output_size in the call operator

Runs the forward pass.